Matrix Algebra

Definitions

Let m and n be positive integers.

- An $m \times n$ matrix is a rectangular array of numbers having m rows and n columns. Such a matrix is said to have size $m \times n$.
- a_{ij} is the (i, j)-entry of a matrix, which is the entry in row i and column j.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]$$

Other keywords: diagonal entries, the main diagonal, identity matrix, zero matrix

Example:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix} = [a_{ij}] \qquad a_{23} = -1$$

the main diagonal

(Zero matrix) $0_{2\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(Identity matrix) $I_4 = I_{4\times 4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Matrix addition

Definition

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $m \times n$ matrices. Then $A + B = C$ where C is the $m \times n$ matrix $C = [c_{ij}]$ defined by

$$c_{ij} = a_{ij} + b_{ij}$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} -1 & 2 \\ 4 & 3 \end{bmatrix} \qquad C = A + B = \begin{bmatrix} 0 & 4 \\ 7 & 7 \end{bmatrix}$$

$$C_{12} = a_{12} + b_{12} = 2 + 2 = 4$$

Theorem (Properties of Matrix Addition)

Let A, B and C be $m \times n$ matrices. Then the following properties hold.

1. $A + B = B + A$ *(matrix addition is commutative).*
2. $(A + B) + C = A + (B + C)$ *(matrix addition is associative).*
3. *There exists an $m \times n$ zero matrix, 0, such that $A + 0 = A$. (existence of an additive identity).*
4. *There exists an $m \times n$ matrix $-A$ such that $A + (-A) = 0$. (existence of an additive inverse).*

Proof of ②:

The (i,j)-entry of $(A+B)+C$ is $(a_{ij}+b_{ij})+c_{ij}$,

which is equal to the (i,j)-entry of $A+(B+C)$, i.e., $a_{ij}+(b_{ij}+c_{ij})$.

Scalar Multiplication

Definition

Let $A = \left[a_{ij}\right]$ be an $m \times n$ matrix and let k be a scalar. Then $kA = \left[ka_{ij}\right]$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad 2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Theorem (Properties of Scalar Multiplication)

Let A, B be $m \times n$ matrices and let $k, p \in \mathbb{R}$ (scalars). Then the following properties hold.

1. $k(A + B) = kA + kB$. *(scalar multiplication distributes over matrix addition).*
2. $(k + p)A = kA + pA$. *(addition distributes over scalar multiplication).*
3. $k(pA) = (kp)A$. *(scalar multiplication is associative).*

$$3\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} -3 & 0 \\ 3 & -3 \end{bmatrix} = (3-1+3)\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = 5\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -5 & 5 \end{bmatrix}$$

Matrix Multiplication

Definition (Product of two matrices)

Let A be an $m \times n$ matrix and let B be an $n \times p$ matrix, whose columns are $B_1, B_2, \ldots, B_p$. The product of A and B is

$$AB = A\begin{bmatrix} B_1 & B_2 & \cdots & B_p \end{bmatrix} = \begin{bmatrix} AB_1 & AB_2 & \cdots & AB_p \end{bmatrix}$$

Question: What is the size of AB?

Problem

Find the product AB of matrices $A = \underbrace{\begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}}_{2\times 3}$ and $B = \underbrace{\begin{bmatrix} -1 & 1 & 2 \\ 0 & -2 & 4 \\ 1 & 0 & 0 \end{bmatrix}}_{3\times 3}$

$$\begin{array}{ccc} \vec{B}_1 & \vec{B}_2 & \vec{B}_3 \end{array}$$

$$A\vec{B}_1 = \begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}_{2\times 3}\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}_{3\times 1} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}_{2\times 1}$$

$$A\vec{B}_2 = \begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}_{2\times 3}\begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}_{3\times 1} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}_{2\times 1}$$

$$A\vec{B}_3 = \begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}_{3\times 1} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}_{2\times 1}$$

$$AB = \begin{bmatrix} 4 & -1 & -2 \\ -1 & 4 & 0 \end{bmatrix}_{2\times 3}$$

Computing the (i, j)-entry of a product

Let $A = [a_{ij}]$ be an $m \times n$ matrix and $B = [b_{ij}]$ be an $n \times p$ matrix. Then the (i, j)-entry of AB is given by

$$a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Example:

$$A = \begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} -1 & 1 & 2 \\ 0 & -2 & 4 \\ 1 & 0 & 0 \end{bmatrix} \qquad C = AB$$

$$C_{23} = a_{21} b_{13} + a_{22} b_{23} + a_{23} b_{33} = 4 - 4 + 0 = 0$$

Problem

Let $A = \begin{bmatrix} 1 & 2 \\ -3 & 0 \\ 1 & -4 \end{bmatrix}_{3\times 2}$ and $B = \begin{bmatrix} 1 & -1 & 2 & 0 \\ 3 & -2 & 1 & -3 \end{bmatrix}_{2\times 4}$

- *Does AB exist? If so, compute it.* AB is defined and its size will be 3x4.
- *Does BA exist? If so, compute it.* BA is not defined, since the number of columns in B is NOT the same as the number of rows in A.

$$AB = \begin{bmatrix} 1 & 2 \\ -3 & 0 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 & 0 \\ 3 & -2 & 1 & -3 \end{bmatrix} = \begin{bmatrix} 7 & -5 & 4 & -6 \\ -3 & 3 & -6 & 0 \\ -11 & 7 & -2 & 12 \end{bmatrix}$$

Problem

Let $A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 \\ 0 & 3 \end{bmatrix}$.

- *Does AB exist? If so, compute it.*
- *Does BA exist? If so, compute it.*

$AB = \begin{bmatrix} -1 & 1 \\ -2 & -1 \end{bmatrix}$ whereas $BA = \begin{bmatrix} 1 & -1 \\ 6 & -3 \end{bmatrix}$.

Note that AB and BA are both defined, however, $AB \neq BA$.

Theorem

Let A, B, and C be matrices of the appropriate sizes, and let $r \in \mathbb{R}$ be a scalar. Then the following properties hold.

1. $A(B+C) = AB + AC$. *(matrix multiplication distributes over matrix addition)*
2. $(B+C)A = BA + CA$. *(matrix multiplication distributes over matrix addition)*
3. $A(BC) = (AB)C$. *(matrix multiplication is associative)*
4. $r(AB) = (rA)B = A(rB)$.

Problem

Let A and B be $m \times n$ matrices, and let C be an $n \times p$ matrix. Prove that if A and B commute with C, then $A+B$ commutes with C.

If $AC = CA$ and $CB = BC$, then

$C(A+B) \overset{1}{=} CA + CB = AC + BC \overset{2}{=} (A+B)C \implies$ C and $A+B$ commute.

Matrix Transposition

Definition

If A is an $m \times n$ matrix, then its transpose, denoted A^T, is the $n \times m$ whose i^{th} row is the j^{th} column of A, $1 \leq i \leq n$; i.e., if $A = [a_{ij}]$, then

$$A^T = [a_{ij}]^T = [a_{ji}]$$

i.e., the (i, j)-entry of A^T is the (j, i)-entry of A.

Example: If $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \\ 1 & 0 \end{bmatrix}_{3\times 2}$, Then $A^T = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix}_{2\times 3}$.

Theorem (Properties of the Transpose of a Matrix)

Let A and B be m × n matrices, C be a n × p matrix, and $r \in \mathbb{R}$ a scalar. Then

1. $(A^T)^T = A$
2. $(rA)^T = rA^T$
3. $(A + B)^T = A^T + B^T$
4. $(AC)^T = C^T A^T$

Problem: A matrix A is called symmetric iff $A^T = A$. Show that if A and B are symmetric then $A^T + 2B$ is symmetric.

If A and B are symmetric, then $A = A^T$ and $B = B^T$.

$$(A^T + 2B)^T \overset{3}{=} (A^T)^T + (2B)^T \underset{2}{\overset{1}{=}} A + 2B^T = A^T + 2B$$

Since $(A^T + 2B)^T = A^T + 2B$, we know that $A^T + 2B$ is symmetric by definition.

Question: Let matrices A and B be symmetric matrices. Is AB symmetric?

Problem

Find the matrix A if $\left(A + 3\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 4 \end{bmatrix}\right)^T = \begin{bmatrix} 2 & 1 \\ 0 & 5 \\ 3 & 8 \end{bmatrix}$.

$$\left(\left(A + 3\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 4 \end{bmatrix}\right)^T\right)^T = \begin{bmatrix} 2 & 1 \\ 0 & 5 \\ 3 & 8 \end{bmatrix}^T$$

$$A + 3\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 \\ 1 & 5 & 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 1 & 5 & 8 \end{bmatrix} - \begin{bmatrix} 3 & -3 & 0 \\ 3 & 6 & 12 \end{bmatrix} = \begin{bmatrix} -1 & 3 & 3 \\ -2 & -1 & -4 \end{bmatrix}$$